Permanental Dominance for Generalized Matrix Functions of Order Four

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Abstract

For a subgroup $H \leq \mathfrak{S}_4$ and an irreducible complex character χ of H, let d_χ^H denote the associated generalized matrix function. We prove that $d_\chi^H(A)/\chi(1) \leq \operatorname{per} A$ for every 4×4 Hermitian positive-semidefinite matrix A. Thus permanental dominance holds in order four for all subgroup characters, not only for the five irreducible immanants associated with $\mathfrak{S}_4$. After reducing to correlation matrices, we classify the thirty-seven irreducible-character cases arising from the eleven conjugacy classes of subgroups of $\mathfrak{S}_4$. Thirty-five cases follow from principal-minor, moment, and block-contraction inequalities. The two remaining conjugate characters of A_4 reduce to a polynomial inequality on the cone of 3×3 positive-semidefinite Gram matrices. Boundary sum-of-squares, an exact interior Hessian calculation, and rational Gram certificates establish its nonnegativity.

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1. Introduction

The determinant and permanent are the two best-known members of the family of generalized matrix functions. Schur's classical theorem places the determinant below every normalized irreducible immanant on the Hermitian positive-semidefinite cone [1]. In the opposite direction, Lieb conjectured that

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the permanent lies above every normalized immanant [2]; the formulation became known as the permanental-dominance conjecture through the work of Merris [3]. In its subgroup form, for an irreducible character χ of $H \leq \mathfrak{S}_n$, the conjectured inequality is

$$\frac{1}{\chi(1)} \sum_{\sigma \in H} \chi(\sigma) \prod_{i=1}^{n} A_{i,\sigma(i)} \leq \operatorname{per} A \qquad (A \succeq 0).$$

Taking $H = \mathfrak{S}_n$ recovers the usual irreducible-immanant statement.

The conjecture has generated a substantial literature on dominance orderings and sufficient analytic mechanisms. Representative advances include the ordering of hook immanants [4], inequalities for generalized matrix functions and row-appending methods [5–7], and the spectral approach of Soules [8]. Broader accounts and historical surveys are given by Cheon and Wanless [9], Zhang [10], and Wanless [11]. More recent work relates immanant inequalities to weight-space methods [12], while the immanant form of permanental dominance has acquired an operational interpretation through generalized bosonic bunching probabilities [13].

Most of the literature concerns the immanant specialization $H = \mathfrak{S}_n$. That specialization is known in several broad families and, in particular, through order thirteen; these results do not by themselves cover characters of proper subgroups [10, 11]. In order four, Barrett, Hall, and Loewy characterized the cone of class-function inequalities and hence the inequalities among linear combinations of $\mathfrak{S}_4$-immanants [14]. James also exhibited a positive-semidefinite matrix giving equality for one of the non-real A_4 characters [9, 11, 15].

These order-four results do not settle the full subgroup problem. A character of a proper subgroup, extended by zero to $\mathfrak{S}_4$, need not be a class function on $\mathfrak{S}_4$; in particular, odd conjugation interchanges the two non-real characters of A_4. The distinction produces a substantially larger finite problem: $\mathfrak{S}_4$ has five irreducible characters, whereas all subgroups of $\mathfrak{S}_4$ contribute thirty-seven irreducible-character cases. To the best of our knowledge, a uniform proof covering every subgroup $H \leq \mathfrak{S}_4$ and every irreducible character of H has not previously been recorded.

This paper proves precisely that all-subgroup statement for every 4×4 Hermitian positive-semidefinite matrix. The proof has three main components. First, diagonal scaling reduces the problem to correlation matrices, and subgroup classification reduces it to a finite list of character cases. Second, general scalar and block inequalities settle all but the two conjugate non-real characters of A_4. Third, the remaining character gap is converted into one polynomial inequality on the cone of 3×3 Gram matrices and proved by exact boundary and interior certificates.

The organizing idea is to write the exceptional gap as

$$c_0(u, v, w) + q^* K(u, v, w) q.$$

The block positivity constraint says that $B(u, v, w) - qq^*$ is a Gram matrix. After dividing out the nonzero coordinates of q, this residual becomes the Gram matrix R of three vectors, and the exceptional gap factors as

$$|q_1 q_2 q_3|^2 \mathcal{P}(R).$$

The problem is therefore moved from a character sum on four indices to the nonnegativity of one explicit polynomial on the cone of 3×3 Gram matrices.

Section 2 fixes the normalization and removes the diagonal parameters. Section 3 proves that the finite list of cases is complete, and Section 4 handles the cases covered by the general inequalities. Sections 5 and 6 contain the geometric argument. The proof is assembled in Section 7; exact coefficient certificates and the Lean verification statement are kept in the appendices so that they do not interrupt the main line of reasoning.

2. Statement and general reductions

Let $n \geq 1$, let $H \leq \mathfrak{S}_n$, and let $\chi : H \to \mathbb{C}$ be a character. For $A = (A_{ij}) \in M_n(\mathbb{C})$ put

$$d_\chi^H(A) := \sum_{\sigma \in H} \chi(\sigma) \prod_{i=1}^{n} A_{i,\sigma(i)}, \qquad \operatorname{per} A := \sum_{\sigma \in \mathfrak{S}_n} \prod_{i=1}^{n} A_{i,\sigma(i)}. \tag{1}$$

The normalization in the permanental-dominance conjecture is $d_\chi^H / \chi(1)$. Notice that no complex conjugate is placed on $\chi(\sigma)$ in (1); this convention agrees with all formulae below.

Lemma 2.1 (Reality). *If $A = A^*$ and χ is the character of a finite-dimensional complex representation of H, then $d_\chi^H(A)$ and $\operatorname{per} A$ are real.*

Proof. Every finite-group representation may be made unitary, and hence $\overline{\chi(\sigma)} = \chi(\sigma^{-1})$. Hermitian symmetry gives

$$\overline{\prod_i A_{i,\sigma(i)}} = \prod_i A_{\sigma(i),i} = \prod_j A_{j,\sigma^{-1}(j)}.$$

Conjugating the sum and replacing σ by σ^{-1} proves the first claim. The second is the special case $H = \mathfrak{S}_n$, $\chi = 1$. $\square$

Theorem 2.2 (Permanental dominance in order four)**.** *Let $H \leq \mathfrak{S}_4$, let V be an irreducible finite-dimensional complex representation of H, and let χ be its character. For every $A \in M_4(\mathbb{C})$ with $A \succeq 0$,*

$$\frac{d_\chi^H(A)}{\dim V} \leq \operatorname{per} A. \tag{2}$$

Theorem 2.2 concerns order four only; it makes no assertion in arbitrary order.

2.1. Reduction to correlation matrices

The normalization step is elementary but important because it removes all diagonal parameters without an invertibility hypothesis.

Lemma 2.3 (Diagonal congruence)**.** *Let $s_1, \ldots, s_n \in \mathbb{R}$, and let*

$$D = \operatorname{diag}(s_1, \ldots, s_n).$$

Then

$$d_\chi^H(DAD) = \left(\prod_i s_i\right)^2 d_\chi^H(A), \qquad \operatorname{per}(DAD) = \left(\prod_i s_i\right)^2 \operatorname{per} A.$$

Proof. For every permutation σ,

$$\prod_i (DAD)_{i,\sigma(i)} = \prod_i (s_i A_{i,\sigma(i)} s_{\sigma(i)}) = \left(\prod_i s_i\right)^2 \prod_i A_{i,\sigma(i)}.$$

Summation proves both identities. $\square$

Lemma 2.4 (Correlation reduction)**.** *It is enough to prove Theorem 2.2 for positive-semidefinite matrices whose diagonal entries are all one.*

Proof. Suppose first that $A_{ii} > 0$ for all i and put $s_i = \sqrt{A_{ii}}$. Since a Hermitian matrix has real diagonal, the matrix

$$C = \operatorname{diag}(s_i^{-1}) A \operatorname{diag}(s_i^{-1})$$

is positive semidefinite and has diagonal one. Moreover,

$$A = \operatorname{diag}(s_i) C \operatorname{diag}(s_i).$$

Thus Lemma 2.3 transfers the desired inequality from C to A.

If $A_{ii} = 0$ for some i, positivity of the 2×2 principal minors implies $A_{ij} = A_{ji} = 0$ for every j. Each permutation monomial then contains a zero entry in row i, so both sides of (2) vanish. □

We therefore write a general 4×4 correlation matrix as

$$A(a,b,c,d,e,f) = \begin{pmatrix} 1 & a & b & c \\ \bar{a} & 1 & d & e \\ \bar{b} & \bar{d} & 1 & f \\ \bar{c} & \bar{e} & \bar{f} & 1 \end{pmatrix} \succeq 0. \tag{3}$$

3. Finite character reduction

3.1. Subgroups and characters

Up to conjugacy, the subgroups of $\mathfrak{S}_4$ have the following eleven types. Here the superscripts $\mathrm{t}, \mathrm{d}, \mathrm{n}$ denote transposition, double transposition, and normal, respectively; in the dihedral row, $r = (1234)$ and $s = (13)$.

type	representative	group	$\lvert H \rvert$	$\lvert \operatorname{Irr}(H) \rvert$
1	$\{1\}$	1	1	1
C_2^{t}	$\langle (12) \rangle$	C_2	2	2
C_2^{d}	$\langle (12)(34) \rangle$	C_2	2	2
C_3	$\langle (123) \rangle$	C_3	3	3
C_4	$\langle (1234) \rangle$	C_4	4	4
V_4^{n}	$\{1, (12)(34), (13)(24), (14)(23)\}$	C_2^2	4	4
V_4^{d}	$\langle (12), (34) \rangle$	C_2^2	4	4
S_3	$\operatorname{Stab}(4)$	S_3	6	3
D_8	$\langle r, s \rangle$	D_8	8	5
A_4	A_4	A_4	12	4
S_4	S_4	S_4	24	5

The final column sums to 37.

Lemma 3.1 (Subgroup classification)**.** *Every subgroup of $\mathfrak{S}_4$ is conjugate to exactly one of the eleven types in the preceding table.*

Proof. An intransitive subgroup preserves its orbit partition. For partition $2+1+1$ it is contained in S_2 and gives 1 or C_2^{t}. For $2+2$ it is contained in $S_2 \times S_2$ and gives, up to conjugacy, $1, C_2^{\mathrm{t}}, C_2^{\mathrm{d}}$, or V_4^{d}. For $3+1$ it is contained in the displayed point stabilizer S_3 and gives $1, C_2^{\mathrm{t}}, C_3$, or S_3. This accounts for every intransitive type.

If the action is transitive, orbit–stabilizer shows that 4 divides $|H|$, while Lagrange's theorem gives $|H| \mid 24$. A transitive group of order four is regular and is either C_4 or the normal double-transposition group V_4^{n}. A group of order eight is a Sylow 2-subgroup, hence is conjugate to $\langle(1234),(13)\rangle \cong D_8$. The cases of order twelve and twenty-four are A_4 and S_4, respectively. The listed types are distinguished by order, orbit structure, and, for the two order-two and two Klein-four types, by cycle type. $\square$

The character data used below are recorded in Appendix Appendix A. Orthogonality of the character-table rows and the identity $\sum_\chi \chi(1)^2 = |H|$ prove that no irreducible character is missing. For the two exceptional A_4 characters we fix

$$\omega = -\frac{1}{2} - \frac{\sqrt{3}}{2} i, \qquad \omega^2 + \omega + 1 = 0. \tag{4}$$

3.2. Conjugation invariance

Lemma 3.2 (Transport)**.** *Let $g \in \mathfrak{S}_4$, $K = g^{-1}Hg$, and let $\chi^g(g^{-1}\sigma g) = \chi(\sigma)$. If $A^g_{ij} = A_{g(i),g(j)}$, then*

$$d^K_{\chi^g}(A^g) = d^H_\chi(A), \qquad \operatorname{per} A^g = \operatorname{per} A, \qquad A \succeq 0 \Longrightarrow A^g \succeq 0.$$

Proof. Simultaneous permutation of rows and columns preserves positivity and the permanent. For a permutation monomial, changing i to $g(i)$ gives

$$\prod_i A^g_{i,(g^{-1}\sigma g)(i)} = \prod_j A_{j,\sigma(j)}.$$

Summing proves the assertion. $\square$

Thus it suffices to prove dominance in the thirty-seven irreducible-character cases associated with the eleven fixed representatives.

4. Character cases from general inequalities

4.1. Four cycle aggregates

For (3) define

$$\mathsf{T} = |a|^2 + |b|^2 + |c|^2 + |d|^2 + |e|^2 + |f|^2, \tag{5}$$
$$\mathsf{D} = |a|^2|f|^2 + |b|^2|e|^2 + |c|^2|d|^2, \tag{6}$$
$$\mathsf{C} = 2\,\mathrm{Re}(ad\bar{b} + ae\bar{c} + bf\bar{c} + df\bar{e}), \tag{7}$$
$$\mathsf{F} = 2\,\mathrm{Re}(adf\bar{c} + ae\bar{f}\bar{b} + b\bar{d}e\bar{c}). \tag{8}$$

Enumerating the cycle types of $\mathfrak{S}_4$ gives

$$\operatorname{per} A = 1 + \mathsf{T} + \mathsf{D} + \mathsf{C} + \mathsf{F}. \tag{9}$$

The five normalized S_4 generalized matrix functions, in the order of the characters in (A.1), are

$$\begin{aligned} p_{[4]} &= 1 + \mathsf{T} + \mathsf{D} + \mathsf{C} + \mathsf{F}, \\ p_{[1^4]} &= 1 - \mathsf{T} + \mathsf{D} + \mathsf{C} - \mathsf{F}, \\ p_{[31]} &= 1 + \tfrac{1}{3}\mathsf{T} - \tfrac{1}{3}\mathsf{D} - \tfrac{1}{3}\mathsf{F}, \\ p_{[211]} &= 1 - \tfrac{1}{3}\mathsf{T} - \tfrac{1}{3}\mathsf{D} + \tfrac{1}{3}\mathsf{F}, \\ p_{[22]} &= 1 + \mathsf{D} - \tfrac{1}{2}\mathsf{C}. \end{aligned} \tag{10}$$

Lemma 4.1 (Scalar correlation inequalities)**.** *If A in* (3) *is positive semidefinite, then*

$$0 \le \mathsf{T}, \quad 0 \le \mathsf{D}, \quad \mathsf{D} \le \frac{\mathsf{T}^2}{4}, \quad -2\mathsf{D} \le \mathsf{F} \le 2\mathsf{D} \le \mathsf{T}, \tag{11}$$
$$\frac{\mathsf{T}^2}{2} \le \mathsf{T} + \frac{3}{2}\mathsf{C}, \qquad 1 + \mathsf{D} \le \operatorname{per} A. \tag{12}$$

In addition, if

$$U = |a|^2 + |b|^2 + |d|^2, \qquad z = ad\bar{b}, \tag{13}$$

then

$$U + 3\,\mathrm{Re}\, z - \sqrt{3}\,\mathrm{Im}\, z \ge 0, \qquad U + 3\,\mathrm{Re}\, z + \sqrt{3}\,\mathrm{Im}\, z \ge 0. \tag{14}$$

Every 3×3 *principal permanent is at most* $\operatorname{per} A$*.*

Proof. The inequalities $|a|^2, \dots, |f|^2 \le 1$ follow from the 2×2 principal minors. The first four estimates in (11) are then AM–GM and $2|\operatorname{Re}(xy)| \le |x|^2 + |y|^2$, paired over opposite edges.

For the moment inequality, let $\lambda_1, \dots, \lambda_4 \ge 0$ be the eigenvalues of A. Direct trace expansion gives

$$\sum \lambda_i = 4, \quad \sum \lambda_i^2 = 4 + 2\mathsf{T}, \quad \sum \lambda_i^3 = 4 + 6\mathsf{T} + 3\mathsf{C}.$$

The exact identity

$$\left(\sum_i \lambda_i\right)\left(\sum_i \lambda_i^3\right) - \left(\sum_i \lambda_i^2\right)^2 = \sum_{i<j} \lambda_i\lambda_j(\lambda_i - \lambda_j)^2$$

implies the first inequality in (12). It also gives $\mathsf{C} \ge (\mathsf{T}^2 - 2\mathsf{T})/3$. To prove the second inequality, subtract $1 + \mathsf{D}$ from (9). If $\mathsf{T} \le 2$, then

$$\mathsf{T} + \mathsf{C} + \mathsf{F} \ge \mathsf{T} + \frac{\mathsf{T}^2 - 2\mathsf{T}}{3} - 2\mathsf{D} \ge \frac{\mathsf{T}(2 - \mathsf{T})}{6} \ge 0.$$

If $\mathsf{T} \ge 2$, the moment bound gives $\mathsf{C} \ge 0$, while $2\mathsf{D} \le \mathsf{T}$ and $\mathsf{F} \ge -2\mathsf{D}$ give the same conclusion. In particular, $1 \le \operatorname{per} A$.

We next prove (14). Put $u = |a|^2$, $v = |b|^2$, $w = |d|^2$, $S = u + v + w$, and write $z = x + iy$. Positivity of the principal 3×3 block gives

$$0 \le u, v, w \le 1, \qquad x^2 + y^2 = uvw, \qquad 1 - S + 2x \ge 0.$$

Also $27uvw \le S^3$ by AM–GM. If $S \le 9/4$, Cauchy–Schwarz gives

$$(3x \pm \sqrt{3}y)^2 \le 12(x^2 + y^2) \le \frac{4}{9}S^3 \le S^2,$$

which proves both inequalities. If $S \ge 9/4$, then $x \ge (S - 1)/2$ and $|y| \le 1$, so

$$S + 3x \pm \sqrt{3}y \ge \frac{5S - 3}{2} - \sqrt{3} > 0.$$

It remains to prove the principal-permanent contraction. Choose Gram vectors $x_1, \dots, x_4$ for A, write $x \odot y = x \otimes y + y \otimes x$, and set

$$y_1 = x_2 \odot x_3, \qquad y_2 = x_1 \odot x_3, \qquad y_3 = x_1 \odot x_2, qquad v = \bar{c}y_1 + \bar{e}y_2 + \bar{f}y_3.$$

The identity

$$\langle x \odot y, z \odot w \rangle = 2\Big(\langle x, z \rangle \langle y, w \rangle + \langle x, w \rangle \langle y, z \rangle\Big)$$

and direct collection of the cycle types give

$$\|v\|^2 = 2\Big(\operatorname{per} A - perA[\{1,2,3\}]\Big).$$

Hence the displayed principal permanent is at most $\operatorname{per} A$; simultaneous relabeling proves the other three cases. $\square$

We also need two symmetric $2+2$ contractions. Put

$$R_+ := 1 + |b|^2 + |e|^2 + \mathsf{D} + 2\operatorname{Re}(adf\bar{c}), \tag{15}$$

$$R_\wedge := (1-|b|^2)(1-|e|^2) + |af - c\bar{d}|^2. \tag{16}$$

Lemma 4.2 (Block contractions)**.** *For a positive-semidefinite correlation matrix,*

$$R_+ \le \operatorname{per} A, \qquad R_\wedge \le \operatorname{per} A, \qquad 1 + |a|^2|f|^2 + |c|^2|d|^2 \le \operatorname{per} A. \tag{17}$$

Proof. First note the exact identity

$$R_+ = (1+|b|^2)(1+|e|^2) + |af + c\bar{d}|^2. \tag{18}$$

Put $r = (1-|b|^2)^{1/2}$ and

$$y_0 = a + \bar{d}b, \quad y_1 = \bar{d}r, qquad g_0 = c + fb, \quad g_1 = fr.$$

If $Y = |y_0|^2 + |y_1|^2$, $G = |g_0|^2 + |g_1|^2$, and $h = \bar{y}_0 g_0 + \bar{y}_1 g_1$, expanding the definitions gives the certificate

$$\operatorname{per} A - R_+ = Y + G + 2\operatorname{Re}(\bar{e}h). \tag{19}$$

Cauchy–Schwarz and $|e| \le 1$ imply $|\operatorname{Re}(\bar{e}h)|^2 \le YG$. Since $Y + G \ge 2\sqrt{YG} \ge 2|\operatorname{Re}(\bar{e}h)|$, the right-hand side of (19) is nonnegative. This proves the first contraction.

For the exterior contraction, take Gram vectors $x_1, \dots, x_4$ for A. Exterior Cauchy–Schwarz applied to $x_1 \wedge x_3$ and $x_2 \wedge x_4$ gives

$$|af - c\bar{d}|^2 \le (1-|b|^2)(1-|e|^2),$$

where the inner product of the two exterior tensors is $af - c\bar{d}$.

We need two consequences of the first contraction. Relabeling its split twice and using (18) yields $(1+|a|^2)(1+|f|^2) \le \operatorname{per} A$ and $(1+|c|^2)(1+|d|^2) \le \operatorname{per} A$. Put

$$x = |a|^2|f|^2, \qquad y = |c|^2|d|^2, \qquad P = \operatorname{per} A.$$

If $x \le y$, then $x \le |c|^2$ and $1 + x + y \le (1 + |c|^2)(1 + |d|^2) \le P$; if $y \le x$, use the other matching bound. This proves the third inequality in (17).

Finally let $p = |b|^2$, $q = |e|^2$, and $s = |af - c\bar{d}|$. The exterior bound gives $s^2 \le (1 - p)(1 - q)$, hence $0 \le s \le 1$. The triangle inequality gives $s \le \sqrt{x} + \sqrt{y}$. The two matching bounds and AM–GM imply

$$(1 + \sqrt{x})^2 \le P, \qquad (1 + \sqrt{y})^2 \le P.$$

Consequently $(1 + s/2)^2 \le P$. Moreover

$$(1 - p)(1 - q) + s^2 \le (1 + s/2)^2,$$

because $p + q - pq \ge 0$ and $s - 3s^2/4 \ge 0$. Therefore $R_\wedge \le P$, completing the proof. □

4.2. Character cases covered by the general inequalities

The following proposition records the finite analytic part of the proof. Its table makes clear that the only cases not covered are the two non-real A_4 characters.

Proposition 4.3 (The thirty-five elementary character cases)**.** *For every representative subgroup in Lemma 3.1, all normalized generalized matrix functions other than those associated with χ_ω and $\bar{\chi}_\omega$ of A_4 are bounded above by* per A*.*

Proof. The cyclic, Klein-four, and S_3 expansions are as follows. In the C_3 and S_3 cases put $z = ad\bar{b}$.

group	normalized generalized matrix functions
1	1
C_2^{t}	$1 \pm \|a\|^2$
C_2^{d}	$1 \pm \|a\|^2\|f\|^2$
C_3	$1 + 2\,\mathrm{Re}\,z, \quad 1 - \mathrm{Re}\,z \pm \sqrt{3}\,\mathrm{Im}\,z$
V_4^{n}	$1 \pm r \pm s \pm t$ with the four character sign patterns,
	$r = \|a\|^2\|f\|^2$, $s = \|b\|^2\|e\|^2$, $t = \|c\|^2\|d\|^2$
V_4^{d}	$(1 \pm \|a\|^2)(1 \pm \|f\|^2)$
S_3	$1 + \|a\|^2 + \|b\|^2 + \|d\|^2 + 2\,\mathrm{Re}\,z,$
	$1 - \|a\|^2 - \|b\|^2 - \|d\|^2 + 2\,\mathrm{Re}\,z, \quad 1 - \mathrm{Re}\,z$

These expressions follow by listing the elements of the corresponding subgroup. We give the inequalities explicitly. Set

$$U = |a|^2 + |b|^2 + |d|^2, \quad z = x + iy, \quad P_3 = 1 + U + 2x.$$

Lemma 4.1 gives $P_3 \le \operatorname{per} A$ and $U + 3x \pm \sqrt{3}y \ge 0$. Thus

$$P_3 - (1 - x \pm \sqrt{3}y) = U + 3x \mp \sqrt{3}y \ge 0,$$

which verifies the C_3 cases; the value $1 + 2x$ is at most P_3 because $U \ge 0$. For S_3, the trivial case is P_3, the sign case differs from P_3 by $-2U$, and the standard case satisfies $P_3 - (1 - x) = U + 3x \ge 0$, obtained by averaging the two triangle bounds.

The trivial and C_2^{t} cases are bounded by P_3 (or by 1 after choosing the negative sign). For C_2^{d}, the positive value is at most $1 + |a|^2|f|^2 + |c|^2|d|^2 \le \operatorname{per} A$, while the negative value is at most one. For V_4^{n}, the all-positive value is $1 + \mathsf{D} \le \operatorname{per} A$ and each other sign pattern is no larger. For V_4^{d}, all four values are no larger than $(1 + |a|^2)(1 + |f|^2) \le \operatorname{per} A$.

For C_4, put

$$r = |b|^2|e|^2, \qquad z = adf\bar{c}.$$

Its four normalized functions are

$$1 + r + 2\operatorname{Re} z, \quad 1 - r - 2\operatorname{Im} z, \quad 1 + r - 2\operatorname{Re} z, \quad 1 - r + 2\operatorname{Im} z. \tag{20}$$

Put $x = |a|^2|f|^2$ and $y = |c|^2|d|^2$. The estimates

$$2|\operatorname{Re} z|,\ 2|\operatorname{Im} z| \le x + y$$

show that the two values containing $+r$ are at most $1+r+x+y = 1+\mathsf{D} \le \operatorname{per} A$. The other two are at most $1 - r + x + y \le 1 + x + y \le \operatorname{per} A$ by (17).

For D_8, write $p = |b|^2$, $q = |e|^2$. The five normalized functions are

$$\begin{aligned} &R_+, \quad (1 + p)(1 + q) - |af + c\bar{d}|^2, \\ &\qquad (1 - p)(1 - q) - |af - c\bar{d}|^2, \quad R_\wedge, \quad 1 - pq. \end{aligned} \tag{21}$$

The first and fourth are (17). By (18), the second differs from R_+ only by replacing $+|af + c\bar{d}|^2$ with $-|af + c\bar{d}|^2$, and hence is no larger. The third and fifth are at most 1, hence at most $\operatorname{per} A$.

For S_4, the sign case follows from $\mathsf{T}+\mathsf{F} \geq \mathsf{T}-2\mathsf{D} \geq 0$. For the two remaining nontrivial estimates, direct subtraction gives

$$\begin{aligned}
3(\operatorname{per} A - p_{[31]}) &= 2\mathsf{T} + 4\mathsf{D} + 3\mathsf{C} + 4\mathsf{F} \geq \mathsf{T}^2 - 4\mathsf{D} \geq 0, \\
2(\operatorname{per} A - p_{[22]}) &= 2\mathsf{T} + 3\mathsf{C} + 2\mathsf{F} \geq \mathsf{T}^2 - 4\mathsf{D} \geq 0.
\end{aligned}$$

Here we used (11)–(12). Finally, $p_{[211]} \leq 1 \leq \operatorname{per} A$ because $\mathsf{F} \leq \mathsf{T}$ and $\mathsf{D} \geq 0$.

Finally, the trivial A_4 case is the average of the trivial and sign S_4 cases, while its three-dimensional case is the average of the [31] and [211] cases. This proves the proposition. □

5. The exceptional A_4 geometry

5.1. The exceptional character gap

It remains to treat the two conjugate non-real characters of A_4. Their common geometric content is isolated below.

Let

$$B(u,v,w) = \begin{pmatrix} 1 & u & v \\ \bar{u} & 1 & w \\ \bar{v} & \bar{w} & 1 \end{pmatrix}. \tag{22}$$

Define

$$c_0(u,v,w) = |u|^2 + |v|^2 + |w|^2 + 2\operatorname{Re}\big((1-\omega)uw\bar{v}\big) \tag{23}$$

and the Hermitian matrix

$$K(u,v,w) = \begin{pmatrix} 1 & (1-\omega^2)u + v\bar{w} & uw + (1-\omega)v \\ (1-\omega)\bar{u} + w\bar{v} & 1 & (1-\omega^2)w + v\bar{u} \\ \bar{u}\bar{w} + (1-\omega^2)\bar{v} & (1-\omega)\bar{w} + u\bar{v} & 1 \end{pmatrix}. \tag{24}$$

Lemma 5.1 (Character expansion)**.** *For the correlation matrix* (3)*,*

$$\operatorname{per} A - d^{A_4}_{\chi_\omega}(A) = c_0(a,b,d) + \begin{pmatrix} \bar{c} & \bar{e} & \bar{f} \end{pmatrix} K(a,b,d) \begin{pmatrix} c \\ e \\ f \end{pmatrix}. \tag{25}$$

For $\bar{\chi}_\omega$, the same formula holds after an odd simultaneous relabeling of the four indices.

Proof. Choose the labeling of C_+ so that $\chi_\omega(C_+) = \omega$. Listing the twelve elements gives the independently checkable expansion

$$\begin{aligned} d^{A_4}_{\chi_\omega}(A) &= 1 + |a|^2|f|^2 + |b|^2|e|^2 + |c|^2|d|^2 \\ &\quad + \omega\Big(e\bar{d}\bar{f} + a d\bar{b} + b f\bar{c} + c\bar{a}\bar{e}\Big) \\ &\quad + \omega^2\Big(d f\bar{e} + a e\bar{c} + b\bar{a}\bar{d} + c\bar{b}\bar{f}\Big). \end{aligned}$$

The four monomials on the last line are the conjugates of those on the preceding line, in the inverse order, so the displayed quantity is real. Subtracting it from (9), using $\omega^2 = -1 - \omega$, and collecting the constant, diagonal, and off-diagonal terms in (c, e, f) yields (25). An odd permutation interchanges the two three-cycle classes and therefore exchanges χ_ω with its conjugate. $\square$

The remaining task is consequently the following four-vector inequality.

Theorem 5.2 (Four-vector inequality)**.** *Suppose*

$$\mathcal{A} = \begin{pmatrix} B(u, v, w) & q \\ q^* & 1 \end{pmatrix} \succeq 0, \qquad q \in \mathbb{C}^3. \tag{26}$$

Then

$$c_0(u, v, w) + q^* K(u, v, w) q \geq 0. \tag{27}$$

The proof is developed in the residual normalization below and in Section 6.

5.2. Residual-vector normalization

There is a useful geometric meaning behind the next change of variables. Realize the block matrix (26) as the Gram matrix of four unit vectors. The vector corresponding to the last row and column determines a common one-dimensional direction. Subtracting the projections of the first three vectors onto that direction leaves three residual vectors, whose Gram matrix is $B - qq^*$. Dividing the residual vectors by the coordinates of the removed projections does not change positivity; it merely produces the scale-free matrix R below. This is the step that turns a four-vector completion problem into a problem on the cone of three-vector Gram matrices.

Assume first that $q_1 q_2 q_3 \neq 0$. The Schur complement of the final diagonal entry in (26) is $B - qq^* \succeq 0$. Congruence by $\mathrm{diag}(\bar{q}_1^{-1}, \bar{q}_2^{-1}, \bar{q}_3^{-1})$ gives the residual Gram matrix

$$R = \begin{pmatrix} n_1 & h_{12} & h_{13} \\ \bar{h}_{12} & n_2 & h_{23} \\ \bar{h}_{13} & \bar{h}_{23} & n_3 \end{pmatrix} \succeq 0, \tag{28}$$

where

$$n_i = \frac{1-|q_i|^2}{|q_i|^2}, \qquad h_{12} = \frac{u}{q_1\bar{q}_2} - 1, \quad h_{13} = \frac{v}{q_1\bar{q}_3} - 1, \quad h_{23} = \frac{w}{q_2\bar{q}_3} - 1. \tag{29}$$

Thus R is the Gram matrix of three residual vectors w_1, w_2, w_3 with $n_i = \|w_i\|^2$ and $h_{ij} = \langle w_i, w_j\rangle$.

We now define the normalized geometric polynomial. Put

$$\begin{aligned}
E &= n_1n_2 + |h_{12}|^2 + n_1n_3 + |h_{13}|^2 + n_2n_3 + |h_{23}|^2,\\
x_{12,13} &= n_1h_{23} + h_{13}\bar{h}_{12},\\
x_{12,23} &= h_{12}h_{23} + n_2h_{13},\\
x_{13,23} &= n_3h_{12} + h_{13}\bar{h}_{23},\\
S &= E + 2\,\mathrm{Re}(x_{12,13} + x_{12,23} + x_{13,23}),\\
S_\omega &= E + 2\,\mathrm{Re}(\omega^2x_{12,13} + \omega x_{12,23} + \omega^2x_{13,23}),\\
\tau &= h_{13}\bar{h}_{12}\bar{h}_{23},\\
\Gamma &= n_3|h_{12}|^2 + n_2|h_{13}|^2 + n_1|h_{23}|^2 + 3\,\mathrm{Re}\,\tau + \sqrt{3}\,\mathrm{Im}\,\tau.
\end{aligned} \tag{30}$$

Finally set

$$\mathcal{P}(R) = 24 + 6\left(n_1+n_2+n_3+2\,\mathrm{Re}(h_{12}+h_{13}+h_{23})\right) + 2S - S_\omega + \Gamma. \tag{31}$$

Lemma 5.3 (Exact normalization identity)**.** *With* (29)*,*

$$c_0(u,v,w) + q^*K(u,v,w)q = |q_1q_2q_3|^2\mathcal{P}(R). \tag{32}$$

Proof. Equations (29) give

$$u = q_1\bar{q}_2(1+h_{12}), \quad v = q_1\bar{q}_3(1+h_{13}), \quad w = q_2\bar{q}_3(1+h_{23}),$$

and $|q_i|^2(1+n_i) = 1$. Let $Q = |q_1q_2q_3|^2$. Dividing first by Q makes every cancellation visible. Direct substitution in (23) gives

$$\begin{aligned}
\frac{c_0(u,v,w)}{Q} &= (1+n_3)|1+h_{12}|^2 + (1+n_2)|1+h_{13}|^2 + (1+n_1)|1+h_{23}|^2\\
&\quad + 2\,\mathrm{Re}\Big((1-\omega)(1+h_{12})(1+h_{23})(1+\bar{h}_{13})\Big).
\end{aligned} \tag{33}$$

For the quadratic part, the diagonal terms and the three upper-triangular entries of K give

$$\begin{aligned}\frac{q^*Kq}{Q} &= (1+n_2)(1+n_3)+(1+n_1)(1+n_3)+(1+n_1)(1+n_2)\\ &\quad + 2\,\mathrm{Re}\Big((1-\omega^2)(1+n_3)(1+h_{12})+(1+h_{13})(1+\bar h_{23})\\ &\qquad + (1+h_{12})(1+h_{23})+(1-\omega)(1+n_2)(1+h_{13})\\ &\qquad + (1-\omega^2)(1+n_1)(1+h_{23})+(1+h_{13})(1+\bar h_{12})\Big). \end{aligned} \tag{34}$$

Equations (33)–(34) are obtained term by term; for example,

$$\frac{|u|^2}{Q} = (1+n_3)|1+h_{12}|^2, qquad \frac{uw\bar v}{Q} = (1+h_{12})(1+h_{23})(1+\bar h_{13}).$$

Expanding these two displayed identities and using $1+\omega+\omega^2=0$ groups the degree-two terms as $2S - S_\omega$ and the degree-three terms as Γ; the remaining terms are the constant and linear part in (31). Thus their sum is $\mathcal{P}(R)$, which proves (32). □

Thus Theorem 5.2 follows from the next result.

Theorem 5.4 (Geometric polynomial)**.** *For every Hermitian matrix R of the form* (28)*,*

$$R \succeq 0 \quad\Longrightarrow\quad \mathcal{P}(R) \geq 0.$$

6. Positivity of the geometric polynomial

The proof uses the geometry of the Gram cone in three stages. On the rank-one locus the three residual vectors are collinear and the polynomial has a direct sum-of-squares form. In the positive-definite region, two residual vectors are used as coordinates and the third is minimized through an explicit two-dimensional Hermitian Hessian. Finally, every singular Gram matrix is obtained as a limit of positive-definite ones. The rank-one calculation is also the algebraic base coefficient in the interior minimum certificate, so it is both explanatory and logically active.

6.1. The rank-one boundary

Suppose first that R has rank one. Write

$$n_1 = |x|^2, \quad n_2 = |y|^2, \quad n_3 = |z|^2, \qquad h_{12} = \bar{x}y, \quad h_{13} = \bar{x}z, \quad h_{23} = \bar{y}z.$$

Define

$$\begin{aligned} R_0(x,y,z) &= 4 + |x+y+z|^2 + |xy+xz+yz|^2 \\ &\qquad - (|xy|^2 + |xz|^2 + |yz|^2) + |xyz|^2, \quad (35) \\ \Phi(x,y,z) &= xy + \omega xz + \omega^2 yz. \quad (36) \end{aligned}$$

Lemma 6.1 (Rank-one sum of squares)**.** *For all $x, y, z \in \mathbb{C}$,*

$$\mathcal{P}(R) = 2\Big(3R_0(x,y,z) + |\Phi(x,y,z)|^2\Big) \geq 0. \quad (37)$$

Proof. Regard R_0 as a quadratic in z. Put

$$K_0 = |1 + \bar{x}y|^2, \quad L_0 = \overline{x+y} + \overline{xy}(x+y), \quad C_0 = 4 + |x+y|^2.$$

Direct expansion gives

$$R_0 = K_0|z|^2 + 2\,\mathrm{Re}(L_0 z) + C_0$$

and the exact discriminant identity

$$C_0K_0 - |L_0|^2 = 4(1 + \mathrm{Re}(\bar{x}y))^2 \geq 0.$$

Completing the square proves $R_0 \geq 0$. A second direct expansion using $1+\omega+\omega^2 = 0$ gives the equality in (37); its right-hand side is nonnegative. $\square$

6.2. The positive-definite region

It remains to treat a positive-definite residual Gram matrix. Write

$$a = n_1 > 0, \quad b = n_2, \quad c = n_3, \quad h_{12} = g = r - is, \quad p = h_{13}, \quad q = h_{23}, \quad (38)$$

and put

$$G = \begin{pmatrix} a & g \\ \bar{g} & b \end{pmatrix}, \qquad \delta = \det G = ab - r^2 - s^2 > 0, \qquad h = \begin{pmatrix} p \\ q \end{pmatrix}. \quad (39)$$

There is a unique $\zeta \in \mathbb{C}^2$ such that $h = G\zeta$. Positivity of R tested on $(-\zeta, 1)$ gives the Schur slack

$$\zeta^* G\zeta \le c. \tag{40}$$

Set

$$\lambda = 6 + a + b + r^2 + s^2 + 5r - \sqrt{3}s, \tag{41}$$
$$P_0 = 24 + 6(a + b + 2r) + ab + r^2 + s^2, \tag{42}$$
$$\ell = \begin{pmatrix} 6 + (2-\omega)g + (2-\omega^2)b \\ 6 + (2-\omega)a + (2-\omega^2)\bar{g} \end{pmatrix}, \tag{43}$$
$$M = \begin{pmatrix} 1+b & 2-\omega+(1-\omega)g \\ 2-\omega^2+(1-\omega^2)\bar{g} & 1+a \end{pmatrix}, \tag{44}$$
$$T = \lambda I_2 + MG. \tag{45}$$

Lemma 6.2 (Quadratic decomposition)**.** *With the preceding notation,*

$$\begin{aligned} \mathcal{P}(R) &= P_0 + 2\,\mathrm{Re}(\ell^* h) + \lambda c + \mathrm{Re}(h^* M h) \\ &= Q(\zeta) + \lambda(c - \zeta^* G\zeta), \end{aligned} \tag{46}$$

where

$$Q(\zeta) = P_0 + 2\,\mathrm{Re}((G\ell)^*\zeta) + \zeta^*(GT)\zeta. \tag{47}$$

Moreover GT is Hermitian.

Proof. Substitution of (38) in (31) gives the first identity. Replacing h by $G\zeta$, using $G = G^*$ and $T = \lambda I + MG$, gives the second. Finally $GT = \lambda G + GMG$ is Hermitian because G and M are Hermitian. □

Lemma 6.3 (Positivity of the scalar part)**.** *Under $a, b \ge 0$ and $r^2 + s^2 \le ab$, one has $\lambda > 0$.*

Proof. Let $\rho = \sqrt{r^2 + s^2}$. Cauchy–Schwarz and AM–GM give

$$5r - \sqrt{3}s \ge -2\sqrt{7}\rho, \qquad a + b \ge 2\rho.$$

Consequently

$$\lambda \ge (\rho - (\sqrt{7} - 1))^2 + 2(\sqrt{7} - 1) > 0.$$

□

The remaining algebra is organized by the Gram defect δ. Define

$$N = \operatorname{Re}\left(P_0 \det T - \ell^* G \operatorname{adj}(T)\ell\right). \tag{48}$$

Lemma 6.4 (Hessian and minimum certificates)**.** *For $a > 0$ and $\delta \geq 0$,*

$$a^3 \operatorname{Re}\det T = D_0 + \delta D_1 + \delta^2 D_2, \tag{49}$$

where

$$\begin{aligned}
D_0 &= a^3\lambda_0 H_0 > 0,\\
D_1 &= a\Big((4+11a)(r^2+s^2) + a(20+a)r\\
&\qquad - \sqrt{3}a(4+3a)s + a(18+10a+3a^2)\Big) > 0, && (50)\\
D_2 &= a(a+1)(a+2) > 0. && (51)
\end{aligned}$$

Here λ_0 denotes (41) with $b = (r^2+s^2)/a$, and

$$H_0 = 2\lambda_0 - 6 + 4(r^2+s^2) > 0.$$

Furthermore

$$a^3 N = N_0 + \delta N_1 + \delta^2 N_2 + \delta^3 N_3, \tag{52}$$

with $N_0 \geq 0$ and $N_1, N_2, N_3 > 0$.

Proof. The determinant identity follows by expanding (45) after $b = (r^2+s^2+\delta)/a$. To make all sign checks explicit, put $\rho = (r^2+s^2)^{1/2}$ and $k = \sqrt{7}-1$. On the boundary $\delta = 0$, the two estimates in Lemma 6.3 give

$$\lambda_0 \geq (\rho-k)^2 + 2k.$$

Consequently

$$H_0 \geq 6\rho^2 - 4k\rho + 6 = 6\left(\rho - \frac{k}{3}\right)^2 + \frac{2+4\sqrt{7}}{3} > 0. \tag{53}$$

This proves $D_0 > 0$.

For D_1, set

$$Q = 4+11a, \quad L_r = a(20+a), \quad L_s = -\sqrt{3}a(4+3a), \quad C = a(18+10a+3a^2).$$

The identity

$$4Q\Big(Q(r^2+s^2)+L_r r+L_s s+C\Big) = (2Qr+L_r)^2+(2Qs+L_s)^2+4QC-L_r^2-L_s^2 \tag{54}$$

reduces positivity to the exact residual

$$4QC - L_r^2 - L_s^2 = 4a(72 + 126a + 94a^2 + 26a^3) > 0.$$

Since $a, Q > 0$, this proves $D_1 > 0$; the assertion for D_2 is immediate.

The expansion (52) is obtained in the same way from (48). On the boundary $\delta = 0$, the first two residual vectors are collinear. Define

$$x_0 = \sqrt{a}, \qquad y_0 = \frac{g}{\sqrt{a}}, \qquad L_0' = x_0\ell_1 + y_0\ell_2,$$

where ℓ_1, ℓ_2 are the components of (43), evaluated at $b = (r^2 + s^2)/a$. Then $|x_0|^2 = a$, $|y_0|^2 = b$, and $\bar{x}_0 y_0 = g$. Lemma 6.1, viewed as a quadratic in the third scalar coordinate, gives $|L_0'|^2 \le P_0H_0$. Direct substitution in (48) now yields the exact boundary coefficient

$$N_0 = a^3\lambda_0(P_0H_0 - |L_0'|^2) \ge 0. \tag{55}$$

The remaining coefficients are explicit. First,

$$N_3 = (a + 2)(a^2 + 6) > 0.$$

Next, collecting the terms in (r, s) gives

$$\begin{aligned} N_2 = 36(r^2 + s^2) &+ a(120 + 84r - 36\sqrt{3}s + 36(r^2 + s^2)) \\ &+ a^2(60 + 24r - 24\sqrt{3}s + 16(r^2 + s^2)) \\ &+ a^3(16 + 2(r^2 + s^2)) + 2a^4. \end{aligned}$$

Thus $N_2 = Q_2(r^2 + s^2) + L_{2r}r + L_{2s}s + C_2$, where

$$\begin{aligned} Q_2 &= 36 + 36a + 16a^2 + 2a^3, \\ L_{2r} &= 84a + 24a^2, \qquad L_{2s} = -\sqrt{3}(36a + 24a^2), \\ C_2 &= 120a + 60a^2 + 16a^3 + 2a^4. \end{aligned}$$

Applying (54) gives the strictly positive residual

$$4Q_2C_2 - L_{2r}^2 - L_{2s}^2 = 16a(a^6 + 16a^5 + 112a^4 + 318a^3 + 588a^2 + 936a + 1080) > 0.$$

Therefore $N_2 > 0$.

For N_1, put $X = r$, $Y = \sqrt{3}s$ and $\nu = X^2 + Y^2/3$. Then

$$N_1 = 36\nu^2 + aB_1 + a^2B_2 + a^3B_3 + a^4B_4 + 6a^5, \tag{56}$$

where

$$\begin{aligned}
B_1 &= \nu(240 + 168X - 72Y + 54\nu),\\
B_2 &= 432 + 624X - 240Y + 492X^2 + 108Y^2 - 168XY\\
&\qquad + (156X - 84Y)\nu + 38\nu^2,\\
B_3 &= 168 + 72X - 168Y + 54X^2 + 46Y^2 - 84XY\\
&\qquad + (20X - 24Y)\nu + 6\nu^2,\\
B_4 &= 60 + 24X - 24Y + 14\nu.
\end{aligned}$$

Appendix Appendix B gives exact certificates for $B_1 \geq 0$, $B_2 > 0$, $B_4 > 0$, and $B_2 + aB_3 + a^2B_4 > 0$. Hence (56) is strictly positive. □

Proposition 6.5 (Positive-definite residual Gram matrices)**.** *If* $R \succ 0$, *then* $\mathcal{P}(R) > 0$.

Proof. Equations (49) and (50) show $\operatorname{Re}\det T > 0$. Direct multiplication gives the exact leading-entry identity

$$(GT)_{11} = aH_0 + \delta(a+1) > 0. \tag{57}$$

Moreover

$$\det(GT) = \det G \det T = \delta \det T.$$

Because GT is Hermitian, its determinant is real; since $\delta > 0$, this also proves $\det T = \operatorname{Re}\det T$. Hence $\det(GT) = \delta \operatorname{Re}\det T > 0$. The two leading principal minors are positive, so the 2×2 Hermitian matrix GT is positive definite.

The critical-point equation for (47) is $GT\zeta = -G\ell$. Since G is invertible, it is equivalent to $T\zeta = -\ell$, and therefore

$$\zeta_0 = -\frac{\operatorname{adj}(T)\ell}{\operatorname{Re}\det T},$$

and completing the Hermitian square gives

$$\min_{\zeta} Q(\zeta) = \frac{N}{\operatorname{Re}\det T}.$$

By (52), $N > 0$ when $\delta > 0$. Therefore $Q(\zeta) > 0$ for all ζ. Lemma 6.3 and the Schur slack (40) now give

$$\mathcal{P}(R) = Q(\zeta) + \lambda(c - \zeta^* G\zeta) > 0.$$

□

6.3. Closure of the Gram cone

Proof of Theorem 5.4. For $R \succeq 0$ and $t > 0$, the matrix $R + tI$ is positive definite, and hence $\mathcal{P}(R + tI) > 0$ by the preceding proposition. Since $\mathcal{P}$ is a polynomial in the real and imaginary parts of the entries, letting $t \downarrow 0$ gives $\mathcal{P}(R) \geq 0$. □

Proof of Theorem 5.2. If all coordinates of q are nonzero, the residual matrix (28) is positive semidefinite. Theorem 5.4 and (32) give (27).

For arbitrary q, choose phases p_i of modulus one, taking $p_i = q_i/|q_i|$ when $q_i \neq 0$ and $p_i = 1$ otherwise. Convexly interpolate the block matrix (26) with the rank-one correlation completion generated by $(p_1, p_2, p_3, 1)$. Its new last column is $q_i(t) = (1 - t)q_i + tp_i$, which is nonzero for $0 < t < 1$. The interpolated matrix remains positive semidefinite, so the already proved case applies. Letting $t \downarrow 0$ and using continuity proves the result. □

6.4. The resulting matrix certificate

The four-vector inequality has a useful matrix consequence. We record it after the direct proof so that it does not obscure the residual-vector argument that establishes it.

Lemma 6.6 (Ellipsoid-to-adjugate certificate)**.** *Let $B, K \in M_m(\mathbb{C})$ with $B \succeq 0$ and $K = K^*$, and let $c \in \mathbb{R}$. If*

$$\begin{pmatrix} B & q \\ q^* & 1 \end{pmatrix} \succeq 0 \quad \Longrightarrow \quad c + q^* K q \geq 0$$

for every $q \in \mathbb{C}^m$, then

$$(\det B)K + c\, \mathrm{adj}(B) \succeq 0. \tag{58}$$

Proof. If $\det B = 0$, the choice $q = 0$ gives $c \geq 0$ and $\mathrm{adj}(B) \succeq 0$, so (58) follows. If $\det B > 0$, then $B \succ 0$ and the block condition is $q^* B^{-1} q \leq 1$. Given $y \neq 0$, set

$$q = \frac{y}{\sqrt{y^* B^{-1} y}}.$$

The hypothesis, multiplied by $y^*B^{-1}y$, says $y^*(K + cB^{-1})y \geq 0$. Hence $K + cB^{-1} \succeq 0$. Multiplication by $\det B > 0$ and the identity $\operatorname{adj}(B) = (\det B)B^{-1}$ prove the claim. □

Applied to (23)–(24), Lemma 6.6 gives the positive-semidefinite certificate

$$(\det B(u,v,w))K(u,v,w) + c_0(u,v,w)\operatorname{adj}(B(u,v,w)) \succeq 0. \tag{59}$$

7. Completion of the proof

We now combine the geometric inequality with the finite character classification.

Proposition 7.1 (The two non-real A_4 characters)**.** *For every 4×4 positive-semidefinite correlation matrix A,*

$$d_{\chi_\omega}^{A_4}(A) \leq \operatorname{per} A, \qquad d_{\bar{\chi}_\omega}^{A_4}(A) \leq \operatorname{per} A.$$

Proof. Write A in the block form (26) with $B = B(a,b,d)$ and $q = (c,e,f)^T$. Lemma 5.1 and Theorem 5.2 give the first inequality. An odd simultaneous relabeling preserves positivity and the permanent and interchanges the two non-real characters, giving the second. □

Proof of Theorem 2.2. By Lemma 2.4, it is enough to consider correlation matrices. Lemma 3.1 and the transport Lemma 3.2 reduce an arbitrary subgroup and irreducible representation to one of the listed character cases. Proposition 4.3 proves thirty-five of them and Proposition 7.1 proves the remaining two. Lemma 2.1 shows that the compared quantities are real. Finally, Lemma 2.3 returns from correlation matrices to the full positive-semidefinite cone. □

Appendix A. Character data

This appendix records the non-cyclic character data used in the finite character reduction. For S_4, use the conjugacy-class ordering

$$1, \quad (12), \quad (12)(34), \quad (123), \quad (1234).$$

Its irreducible character table is

$$\begin{array}{c|ccccc} & 1 & (12) & (12)(34) & (123) & (1234) \\ \hline [4] & 1 & 1 & 1 & 1 & 1 \\ [1^4] & 1 & -1 & 1 & 1 & -1 \\ [31] & 3 & 1 & -1 & 0 & -1 \\ [211] & 3 & -1 & -1 & 0 & 1 \\ [22] & 2 & 0 & 2 & -1 & 0 \end{array} \tag{A.1}$$

For A_4, let C_+ and C_- be the two conjugacy classes of three-cycles. With ω as in (4), the table is

$$\begin{array}{c|cccc} & 1 & (12)(34) & C_+ & C_- \\ \hline 1 & 1 & 1 & 1 & 1 \\ \chi_\omega & 1 & 1 & \omega & \omega^2 \\ \bar{\chi}_\omega & 1 & 1 & \omega^2 & \omega \\ \chi_3 & 3 & -1 & 0 & 0. \end{array} \tag{A.2}$$

For $D_8 = \langle r, s : r^4 = s^2 = 1, srs = r^{-1}\rangle$, the four linear characters take independent signs on r and s, while its two-dimensional character is $2, -2, 0, 0, 0$ on

$$1, \quad r^2, \quad \{r, r^3\}, \quad \{s, r^2 s\}, \quad \{rs, r^3 s\}.$$

The standard character tables of C_m, C_2^2, and S_3 supply the remaining cases listed in Section 4.

Appendix B. Exact scalar and Gram certificates

This appendix records the non-obvious polynomial certificates used in Lemma 6.4. They contain no numerical approximation. Recall $\nu = X^2 + Y^2/3$.

Appendix B.1. Direct square completions

The coefficient B_2 has the exact decomposition

$$\begin{aligned} B_2 = 38\left(\nu + \frac{39}{19}X - \frac{21}{19}Y + \frac{3}{2}\right)^2 + \frac{448}{19}\left(Y + \frac{3}{32}X - \frac{1083}{448}\right)^2 \\ + \frac{3483}{16}\left(X + \frac{2137}{2322}\right)^2 + \frac{131797}{5418} > 0. \end{aligned}$$

Likewise

$$B_3 + 252 = 6\left(\nu + \frac{5}{3}X - 2Y - \frac{1}{2}\right)^2 + 24\left(Y - \frac{11}{12}X - \frac{15}{4}\right)^2 + \frac{139}{6}\left(X - \frac{249}{139}\right)^2 + \frac{1851}{278} > 0.$$

Further elementary completions give

$$B_1 \geq \frac{112}{3}\nu \geq 0, \qquad B_4 \geq \frac{132}{7} > 0.$$

Appendix B.2. The conditional middle-coefficient certificate

Set

$$\Sigma = 4B_2B_4 + 252B_3.$$

An exact Gram decomposition is

$$\Sigma = 2128\nu\left(\nu + \frac{387}{133}X - \frac{261}{133}Y + \frac{77}{20}\right)^2 + \frac{1}{59850}m^TQm, \tag{B.1}$$

where $m = (1, Y, Y^2, X, XY, X^2)^T$ and

$$Q = \begin{pmatrix} 8739057600 & -4231634400 & 478800000 & 6265576800 & -1795500000 & 47880000 \\ -4231634400 & 2519799114 & -385003080 & -2526627600 & 807975000 & 239400000 \\ 478800000 & -385003080 & 79782560 & 281630160 & -81252000 & -35910000 \\ 6265576800 & -2526627600 & 281630160 & 10931106942 & -2359670040 & 2303554680 \\ -1795500000 & 807975000 & -81252000 & -2359670040 & 685371360 & -243756000 \\ 47880000 & 239400000 & -35910000 & 2303554680 & -243756000 & 1122611040 \end{pmatrix}.$$

Exact rational LDL^T elimination has positive pivots

$$8739057600, \quad \frac{79557301866}{169}, \quad \frac{1112696849800960}{299087601}, \quad \frac{26240992068283038599949}{869294413907}, \quad \frac{25313089709610284699200080}{115380521779374043}, \quad \frac{1406694633431943161674826341030}{10547120712337618624667}.$$

Thus $Q \succ 0$ and $m^TQm > 0$ because the first coordinate of m is one. Equation (B.1) proves $\Sigma > 0$.

If $B_3 \geq 0$, then $B_2 + aB_3 + a^2B_4 > 0$ is immediate. If $B_3 < 0$, the identity

$$4B_2B_4 - B_3^2 = (4B_2B_4 + 252B_3) + (-B_3)(B_3 + 252)$$

shows that the discriminant of the quadratic $B_2 + aB_3 + a^2B_4$ is negative. Hence that quadratic is positive for every $a > 0$. This completes the proof that $N_1 > 0$.

Appendix C. Formal verification

The theorem has been formalized in Lean 4.19.0 with Mathlib 4.19.0. The formal development verifies the subgroup and character enumeration, all cycle expansions, the residual normalization identity, the rational Gram certificate, and the final passage from correlation matrices to the full positive-semidefinite cone. The public theorem is `normalized_permanental_dominance_n4` in the module `PermanentalDominance.N4.NormalizedPermanentalDominance`. A detailed module map and an axiom audit are included with the source. The formalization is supplementary: the identities and sign certificates needed for the mathematical proof have been displayed above rather than delegated to the proof assistant.

OpenAI ChatGPT and Codex were used as AI-assisted tools during mathematical exploration and the development of portions of the Lean formalization. The generated suggestions and code were not treated as mathematical evidence: the stated results were checked by Lean and Mathlib, the imported dependency closure was subjected to the accompanying axiom audit, and the author reviewed the final formal and manuscript content.

The formal source used here was extracted from the immutable merge commit `c5e4d8c61f08` and is mirrored in the companion repository.

Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

No external data were used in this study. The accompanying Lean formalization is available from the companion repository, and the LAA manuscript source is available from its submission branch. The immutable formalization provenance is the commit identified in the preceding section.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI ChatGPT and Codex to assist with mathematical exploration, development of the Lean formalization, literature organization, manuscript drafting, language editing, and LaTeX preparation. The research-method use is described in the formal-verification section above. The author reviewed and edited all generated content and takes full responsibility for the accuracy and integrity of the work.

References

[1] I. Schur, Über endliche gruppen und hermitesche formen, Mathematische Zeitschrift 1 (2–3) (1918) 184–207. doi:10.1007/BF01465095.

[2] E. H. Lieb, Proofs of some conjectures on permanents, Journal of Mathematics and Mechanics 16 (2) (1966) 127–134. doi:10.1512/iumj.1967.16.16008.

[3] R. Merris, The permanental dominance conjecture, in: R. Grone, F. Uhlig (Eds.), Current Trends in Matrix Theory, North-Holland, New York, 1987, pp. 213–223.

[4] P. Heyfron, Immanant dominance orderings for hook partitions, Linear and Multilinear Algebra 24 (1) (1988) 65–78.

[5] T. H. Pate, Partitions, irreducible characters, and inequalities for generalized matrix functions, Transactions of the American Mathematical Society 325 (2) (1991) 875–894. doi:10.1090/S0002-9947-1991-0998356-3.

[6] T. H. Pate, Row appending maps, Ψ-functions, and immanant inequalities for hermitian positive semi-definite matrices, Proceedings of the London Mathematical Society 76 (2) (1998) 307–358. doi:10.1112/S0024611598000100.

[7] T. H. Pate, Tensor inequalities, ζ-functions and inequalities involving immanants, Linear Algebra and its Applications 295 (1–3) (1999) 31–59. doi:10.1016/S0024-3795(99)00014-2.

[8] G. W. Soules, An approach to the permanental-dominance conjecture, Linear Algebra and its Applications 201 (1994) 211–229. doi:10.1016/0024-3795(94)90117-1.

[9] G.-S. Cheon, I. M. Wanless, An update on Minc's survey of open problems involving permanents, Linear Algebra and its Applications 403 (2005) 314–342. doi:10.1016/j.laa.2005.02.030.

[10] F. Zhang, An update on a few permanent conjectures, Special Matrices 4 (1) (2016) 305–316. doi:10.1515/spma-2016-0030.

[11] I. M. Wanless, Lieb's permanental dominance conjecture, in: The Physics and Mathematics of Elliott Lieb: The 90th Anniversary, Vol. 2, EMS Press, Berlin, 2022, pp. 501–516. doi:10.4171/90-2/48.

[12] N. Jing, Y. Liu, J. Zhang, Immanant inequalities and weight spaces (2025). arXiv:2508.20382, doi:10.48550/arXiv.2508.20382.

[13] S. Geller, E. Knill, Measuring multiparticle indistinguishability with the generalized bunching probability, Physical Review A 113 (4) (2026) 042606. doi:10.1103/1x5l-jd26.

[14] W. Barrett, H. T. Hall, R. Loewy, The cone of class function inequalities for the 4-by-4 positive semidefinite matrices, Proceedings of the London Mathematical Society 79 (1) (1999) 107–130. doi:10.1112/S0024611599011909.

[15] G. D. James, Immanants, Linear and Multilinear Algebra 32 (3–4) (1992) 197–210. doi:10.1080/03081089208818163.